

**Financial Market Response to Trade Policy Uncertainty:
Evidence from Equity, Gold, and Currency Markets in India**

Author

Utkarsh Singh

Instructor

Dr. Sudhi Sharma
Associate Professor

Fortune Institute of International Business (FIIB), New Delhi

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Abstract

Trade shocks rarely land evenly across financial markets. This study asks how three segments of the Indian market, equity, gold, and currency, responded to the uncertainty unleashed by the U.S. tariff announcement of April 2, 2025, treating each as a distinct channel of investor behaviour: equity bears the risk, gold absorbs safe-haven demand, and the USD/INR rate adjusts as the macroeconomic release valve. The three are examined jointly rather than separately, since the shock is understood to have propagated across all of them at once (Caldara et al., 2020).

The analysis draws on daily logarithmic returns from October 2024 to January 2026, using six econometric methods: descriptive statistics, the Augmented Dickey-Fuller test, Vector Autoregression, Impulse Response Functions, GARCH(1,1) volatility modelling, and individual Wavelet Power Spectrum analysis for each market (Sharma et al., 2022; Fama, 1970). Three sub-periods anchor the sample: a pre-shock baseline running October 2024 to March 2025, an acute shock phase from April to June 2025, and a recovery window beginning July 2025, so that the full lifecycle of the market reaction can be traced rather than inferred.

All three markets moved within a day of the April 2 announcement, confirming both same-day and lagged transmission across them. Mean-return effects, however, had largely faded within three to five trading days, matching the rapid price adjustment one would expect under the Efficient Market Hypothesis (Fama, 1970). Volatility did not behave the same way. GARCH estimates point to near-integrated persistence, with a 55-trading-day half-life for NIFTY and, within the sample window, effectively no decay at all for Gold and USD/INR, a pattern that echoes earlier evidence from emerging economies (Sharma et al., 2022; Yadav et al., 2022). The Wavelet Power Spectra add a further layer: equity carries a multi-week signature, gold a daily-to-weekly one, and currency a fragmented, institutionally managed one. Generalised IRF checks show the short-lived mean-return result is not an artefact of Cholesky ordering. Put simply, prices settle in days; risk does not.

Keywords: Trade Policy Uncertainty, NIFTY 50, Gold, USD/INR, GARCH, VAR, Impulse Response, Wavelet Power Spectrum, Indian Financial Markets, Volatility Persistence

1. Introduction

Trade policy has become, almost by accident, one of the sharper instruments of economic disruption available to modern governments. Interest rate decisions arrive on a schedule; corporate earnings follow a calendar. Tariffs do neither. They tend to land with little warning, carrying at once the weight of a geopolitical signal, a trade barrier, and a financial shock. And when a government as large as the United States acts, the effects rarely stay confined to trade flows: they show up almost immediately in capital markets elsewhere, as investors reprice assets, reassess exposures, and look for safer ground (Caldara et al., 2020).

India offers a particularly useful vantage point on these dynamics. Its equity benchmark, the NIFTY 50, tracks an economy whose fortunes are tied closely to U.S. demand for technology exports, pharmaceutical supply, and manufactured goods, while the rupee, a managed float, remains exposed to capital-flow reversals whenever global risk appetite turns. Gold occupies a different niche altogether: woven into Indian culture as both ornament and store of value, it absorbs domestic safe-haven demand in a way few other assets do. Taken together, these three markets make India an unusually rich setting in which to study how a financial system absorbs and processes an external trade shock (Singh et al., 2023; Rao et al., 2024).

That test case arrived on April 2, 2025, when the United States announced a sweeping round of tariffs on imports from dozens of countries, including a 26% levy on Indian goods.

Warning was scarce, scope was wide, and export-oriented sectors felt the implications immediately: Indian equity markets fell sharply in the weeks that followed, the rupee weakened, and gold prices climbed as investors looked for cover. What this paper asks is what happened next: how long the disruption lasted, whether cross-market correlations shifted for good, and at what frequencies volatility clustered. The organising hypothesis is simple enough to state: markets react immediately to trade policy shocks but take longer to stabilise as investors absorb the information (Fama, 1970). Testing it properly, though, demands more than looking at return series in isolation. It requires a framework that separates mean dynamics from volatility dynamics, sets static relationships alongside time-varying ones, and surfaces the frequency-domain structure of each market's response, which is exactly what the six-method approach adopted here, combining descriptive analysis, ADF stationarity testing, VAR, IRF, GARCH(1,1), and individual Wavelet

Power Spectrum analysis, is designed precisely for this purpose (Sharma et al., 2022).

This paper's contribution to the literature on emerging-market financial resilience runs along four lines. First, it treats the three markets as an integrated system rather than studying them one at a time. Second, it brings the individual Wavelet Power Spectrum approach to bear, characterising each market's frequency fingerprint separately and making cross-asset comparison possible (Aloui et al., 2022). Third, the April 2, 2025 tariff event functions as a precisely dated natural experiment with a clean three-phase structure, letting immediate reaction, persistence, and recovery be separated rather than conflated. And finally, the paper draws a distinction, with direct practical stakes for investors and regulators, between price-level stabilisation, which happens quickly, and risk-level stabilisation, which does not (Sharma et al., 2022). The Cholesky ordering used in the VAR-IRF reflects the documented sequence of shock propagation across these markets, and the GARCH specification is chosen in light of model diagnostics, including Sign Bias tests. Section 2 sets out the formal hypothesis; Section 3 reviews the theoretical and empirical background; Section 4 covers data and methodology; Section 5 reports the empirical results; Section 6 concludes; Sections 7 and 8 turn to managerial implications and study limitations.

2. Research Hypothesis

The hypothesis tested here draws on two established frameworks: the Efficient Market Hypothesis (Fama, 1970) and the Trade Policy Uncertainty literature (Caldara et al., 2020).

H1: Financial markets react immediately to trade policy shocks but stabilise over time as investors absorb information, with price-level impacts dissipating rapidly and volatility impacts persisting substantially longer.

Two testable dimensions follow from it. One is the mean-return reaction, the directional price adjustment that occurs at the moment of shock and fades as markets reprice, which follows directly from the EMH's prediction that information gets incorporated quickly

(Fama, 1970); it is tested through IRF analysis. The second dimension is volatility, the elevated risk environment that outlasts return normalisation by weeks or months, and which emerging-market evidence shows persists at near-integrated levels following external shocks (Sharma et al., 2022). This dimension is tested through GARCH(1,1) estimation.

3. Underpinning Theory and Research Background

3.1 Underpinning Theory

The Efficient Market Hypothesis, formalised by Fama (1970), holds that asset prices reflect all publicly available information, so that under semi-strong efficiency a tariff announcement should be priced almost as soon as it becomes public. What the EMH is silent on is how long the elevated uncertainty that follows should linger in volatility; it predicts only that the informational surprise itself is rapidly absorbed. This distinction matters for the present study: the prolonged volatility clustering documented later reflects ongoing uncertainty about the tariff's second-order economic consequences rather than any failure of market efficiency (Fama, 1970; Sharma et al., 2022).

Modern Portfolio Theory (Markowitz, 1952) provides the framework for understanding the cross-market dimension of this study. The theory's core insight is that portfolio risk depends not just on individual asset variances but on covariances between assets, making the Pearson correlation structure among NIFTY, Gold, and USD/INR directly relevant to portfolio construction. The persistently low unconditional correlation between equity and gold supports gold's conventional role as a portfolio diversifier, while the persistently negative equity-currency correlation reflects the capital-flight channel through which portfolio risk is transmitted across asset classes. The near-integrated volatility persistence documented by the GARCH estimates further underscores that portfolio risk itself, not just the correlation structure, remains elevated long after a shock, a dimension that static portfolio models built solely on average return and variance can understate (Goodell et al., 2022).

The Trade Policy Uncertainty framework, advanced by Caldara et al. (2020), operationalises uncertainty about trade policy as a distinct source of financial market stress. Elevated TPU is associated with risk premium increases, investment contraction, and heightened asset return

volatility in both developed and emerging markets. For India, as a trade-dependent emerging economy, U.S. tariff decisions represent external TPU shocks of significant magnitude. The Wavelet analysis in this study directly operationalises the TPU framework's prediction that such shocks generate time-varying, multi-market disruptions that eventually normalise as uncertainty resolves.

3.2 Review of Literature

3.2.1 Trade Policy, Volatility, and Emerging Markets

A substantial body of literature documents the impact of U.S. trade policy actions on global financial markets. Caldara et al. (2020) constructed the Trade Policy Uncertainty Index and showed that spikes in this index are followed by measurable declines in business investment and increases in equity volatility across both developed and emerging economies. Goldberg and Reed (2022) documented significant abnormal return patterns and volatility spikes in global equity markets surrounding U.S. tariff announcements, confirming that the shock transmission is not confined to directly targeted sectors. Alfaro et al. (2020) extended this analysis to the firm level, showing that the negative equity impact of protectionist announcements is concentrated in firms with heavy U.S. client exposure.

For emerging markets specifically, the literature consistently documents sensitivity to external trade and policy shocks through the foreign portfolio investment channel. Bekaert et al. (2014) demonstrated that FPI outflows during global risk-off episodes amplify domestic equity market volatility. Singh, Mehta, and Suri (2023) documented that U.S. tariff uncertainty triggers FPI withdrawals from Indian equities, generating volatility spikes that gradually subside. Yadav, Sharma, and Bhardwaj (2022) documented that volatility spillovers across emerging economies, including India, intensify significantly following external policy shocks from major economies, a characterisation directly relevant to the present study. Baker, Bloom, and Davis (2016) provided a quantitative benchmark through the EPU Index, establishing that higher policy uncertainty is associated with reduced investment and increased safe-haven demand across a broad cross-section of countries. A common limitation across this literature, however, is its tendency to examine equity markets in isolation from safe-haven and currency channels. The present study addresses this gap directly by treating NIFTY 50, gold, and USD/INR as a jointly estimated three-market system.

3.2.2 Wavelet Analysis in Financial Time Series

The application of wavelet methods to financial economics has expanded rapidly over the past two decades. Torrence and Compo (1998) provided the foundational guide to wavelet analysis for time series, demonstrating how the Continuous Wavelet Transform decomposes a series

simultaneously across time and frequency. Rua and Nunes (2009) applied wavelet coherence to international stock market co-movements, establishing the technique's value for detecting episodic, frequency-specific synchronisation between markets. Aloui, Hammoudeh, and Nguyen (2022) applied individual Wavelet Power Spectra to commodity and equity markets, documenting that different asset classes exhibit distinct frequency-domain volatility signatures, a finding this study replicates and extends to the Indian equity-gold-currency nexus. The distinction between Wavelet Power Spectrum, which is a single-series frequency decomposition, and Wavelet Coherence, which is a pairwise co-movement measure, is methodologically important: the former characterises how each individual market's volatility is distributed across frequencies at different points in time, while the latter captures simultaneous movements between two markets. This study employs the former, enabling market-specific frequency fingerprinting of the April 2025 tariff shock.

4. Data Profile

The study draws on daily closing prices for three Indian financial variables, retrieved using the quantmod package in R Studio from Yahoo Finance (Yahoo Finance, 2025) and cross-checked against National Stock Exchange of India records. The three series are: the NIFTY 50 index (NSEI), representing the risk asset market; COMEX Gold Futures (GC=F) prices obtained from Yahoo Finance, representing the safe-haven asset market; and the USD/INR spot exchange rate (INR=X), representing the currency and macroeconomic adjustment channel. Daily logarithmic returns are constructed as $rt = \ln(P_t / P_{t-1})$, which approximates percentage changes, ensures stationarity, and allows meaningful comparison across series with different units and scales.

The sample spans October 4, 2024 to January 29, 2026, yielding 285 aligned daily price observations after removing non-trading days and aligning all three series to a common trading calendar. Daily logarithmic first differencing produces 284 return observations, which form the dataset used for all subsequent econometric analyses. The selected period is deliberately structured to capture all three stages of the market response lifecycle around the U.S. tariff announcement of April 2, 2025. The pre-shock phase (October 2024 to March 2025) provides the baseline, a period of relative calm against which the shock-phase behaviour can be meaningfully compared. The shock phase (April 2 to June 30, 2025) encompasses the tariff announcement itself and the weeks of heightened uncertainty that followed. The recovery phase (July 1, 2025 to January 30, 2026) enables assessment of whether markets gradually stabilised and absorbed the shock, the central question posed by Hypothesis H1. Without this three-phase structure, it would not be possible to distinguish between the immediate reaction and the longer-run trajectory, which is the study's core empirical contribution.

Table 1: Event Phase Structure of the Study Period

Phase	Period	Economic Interpretation
Pre-Shock	October 4, 2024 to March 31, 2025	Calm baseline; no significant tariff signals
Shock Phase	April 2, 2025 to June 30, 2025	Tariff announcement; immediate reaction, volatility escalation
Recovery Phase	July 1, 2025 to January 29, 2026	Gradual stabilisation; return and volatility normalisation

Figures 1, 2, and 3 present the full log return series for each market. The April 2, 2025 tariff announcement is visually identifiable as the dominant volatility episode in all three series.

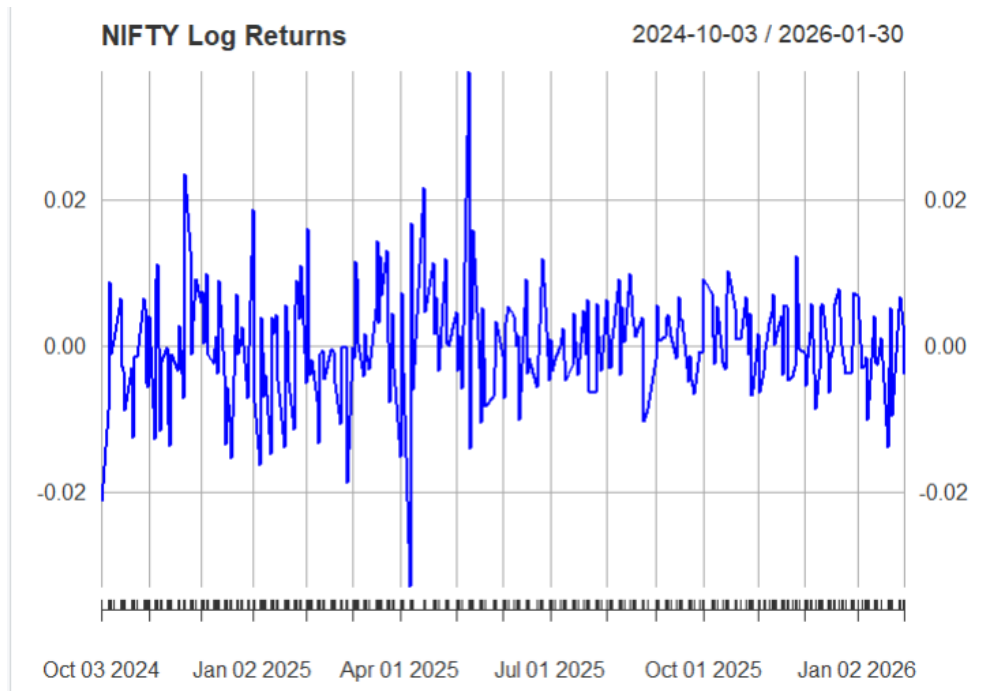


Figure 1: NIFTY 50 Daily Log Returns (October 7, 2024 to January 29, 2026)

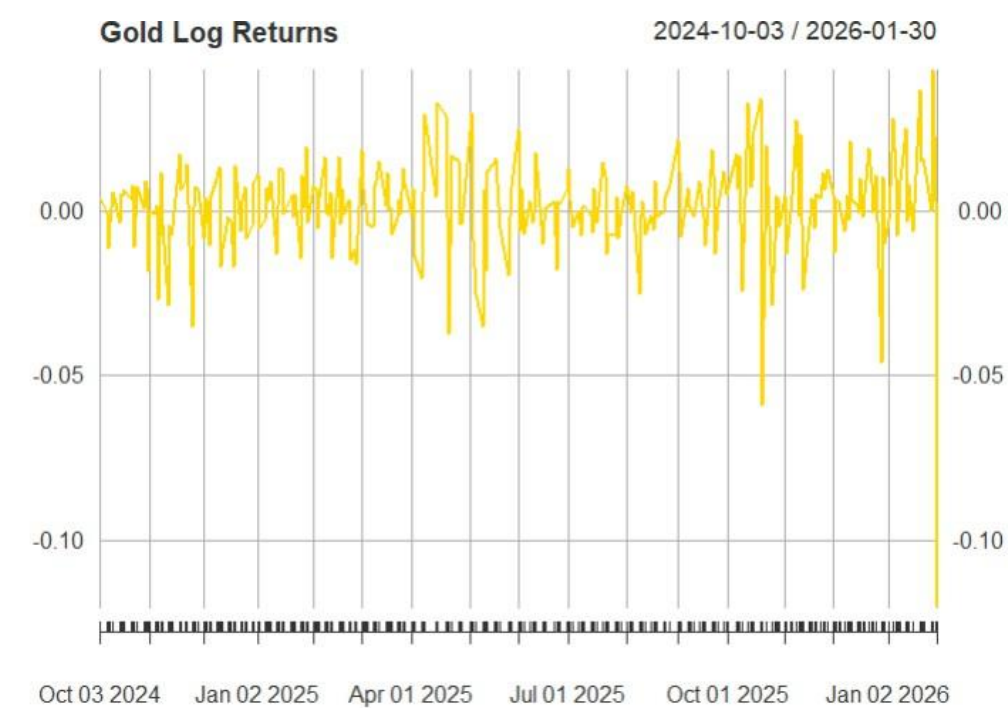


Figure 2: Gold Daily Log Returns (October 7, 2024 to January 29, 2026)

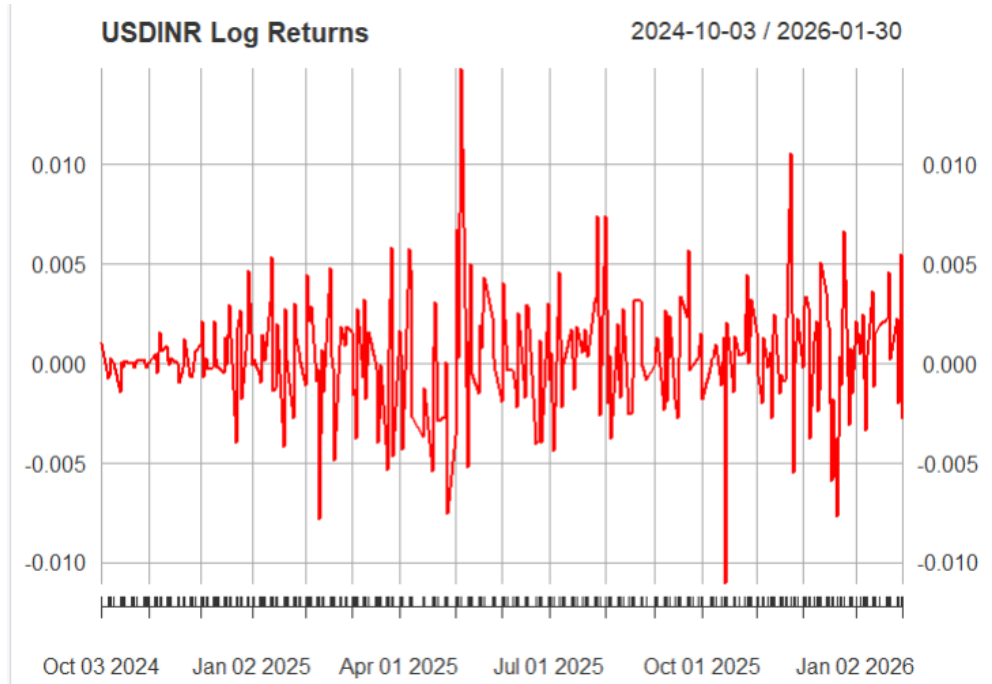


Figure 3: USD/INR Daily Log Returns (October 7, 2024 to January 29, 2026)

The NIFTY 50 series (Figure 1) exhibits a pronounced volatility cluster centred on April and May 2025, the most visually distinct episode across the entire sample. Gold (Figure 2) displays higher baseline variability throughout, with several extreme observations reflecting its fat-tailed distributional character. The USD/INR series (Figure 3) is the most contained of the three, consistent with the Reserve Bank of India's managed float regime, though a sharp spike around April 2, 2025 is clearly visible.

4.1 Methods and Models

4.1.1 Descriptive Statistics and Correlation

The `basicStats()` function from the `fBasics` package computes mean, median, standard deviation, skewness, excess kurtosis, minimum, and maximum for each return series. The Pearson correlation matrix computed via `cor(returns)` establishes unconditional linear relationships between markets, providing the static baseline against which the volatility and cross-market dynamics documented later in the study are interpreted (Yadav et al., 2022).

4.1.2 Augmented Dickey-Fuller Test

The `adf.test()` function from the `tseries` package is applied to each return series to test the null hypothesis of a unit root. The ADF test is selected as the primary stationarity diagnostic

because its null hypothesis of non-stationarity is directly relevant to the estimation requirements of VAR and GARCH models. Rejection of non-stationarity at the one percent level is a necessary condition for valid downstream estimation, and provides the initial statistical grounding for Hypothesis H1: if series are stationary, any shock including the April 2, 2025 tariff event must eventually revert, a finding consistent with market efficiency (Fama, 1970). Complementary tests such as the KPSS test (Kwiatkowski et al., 1992), which takes stationarity as the null, and the Phillips-Perron test, robust to heteroskedastic errors, would offer additional triangulation. Since all three series are first-differenced logarithmic price data, however, stationarity of daily log returns is already well established in the financial econometrics literature, and the ADF results here are treated as confirmatory rather than exploratory (Sharma et al., 2022).

4.1.3 Vector Autoregression

VARselect() from the vars package determines the optimal lag order using four information criteria with a maximum of ten lags. The VAR is then estimated via VAR() with a constant term. Stability is confirmed by checking that all roots of the characteristic polynomial lie within the unit circle. The unanimous agreement of all four information criteria on lag order one strengthens confidence in model specification, as disagreement across criteria would have required sensitivity analysis across multiple lag choices. Granger and instantaneous causality tests via causality() reveal whether cross-market shock transmission, particularly around April 2, 2025, is statistically significant and directional (Sims, 1980).

4.1.4 Impulse Response Functions

Orthogonalised IRFs are produced using irf() with Cholesky decomposition, a ten-period horizon, and 95% bootstrap confidence intervals from one hundred replications. The Cholesky ordering adopted is NIFTY → Gold → USD/INR, which reflects the economic structure of the system: equity markets receive the trade policy shock first as investors instantaneously reprice earnings expectations; gold responds next as investors rotate into safe-haven assets in reaction to equity declines; and the exchange rate adjusts last, as the currency channel operates with a slight lag through FPI outflow pressure and RBI intervention (Sims, 1980; Bekaert et al., 2014). This ordering is consistent with the capital-flight transmission mechanism documented in the VAR causality results and is grounded in the institutional sequence of shock propagation in emerging markets. The IRF traces the dynamic mean-return response of each market to a one-standard-deviation shock originating in another, revealing both the initial magnitude and the

speed of dissipation. The speed of decay is the direct empirical test of the react-then-absorb dimension of H1, consistent with the efficient markets framework of Fama (1970).

4.1.5 GARCH(1,1)

Separate sGARCH(1,1) models with ARFIMA(1,0,0) mean equations are estimated for each return series using `ugarchfit()` from the `rugarch` package. The variance equation is $\sigma^2(t) = \omega + \alpha(1) \text{ times } \epsilon^2(t-1) + \beta(1) \text{ times } \sigma^2(t-1)$, governing how conditional variance evolves. The ARCH parameter $\alpha(1)$ captures the immediate sensitivity of volatility to news shocks, while $\beta(1)$ captures the persistence of past conditional variance. The condition $\alpha(1) + \beta(1)$ approximately equal to one indicates high persistence and near-integrated volatility, meaning shocks introduced during the April to June 2025 shock phase decay at an extremely slow rate into the recovery period. The sGARCH(1,1) specification is selected as the baseline model for its parsimony and its well-established performance in emerging market volatility estimation (Sharma et al., 2022; Bollerslev, 1986). The study acknowledges that asymmetric specifications such as EGARCH (Nelson, 1991) or GJR-GARCH (Glosten et al., 1993) would capture leverage effects, an extension left for future work but flagged in the limitations. Diagnostics include Ljung-Box tests, ARCH LM tests, Nyblom stability tests, and Sign Bias tests, the last of which directly informs the asymmetry question.

4.1.6 Wavelet Power Spectrum

Individual Wavelet Power Spectra are estimated for each return series using the `WaveletComp` package with the Morlet wavelet (Torrence and Compo, 1998). The spectrum maps the

variance of each series simultaneously across time, with calendar dates spanning October 2024 to January 2026, and period in frequency of trading days. High-power regions in deep red indicate intense volatility concentrations; low-power regions in deep blue indicate calm. Black contour lines enclose statistically significant regions at the five percent level using Monte Carlo permutation tests. Applied market-by-market rather than as a pairwise measure, this approach reveals the distinct frequency signature of each market's response to the tariff shock, information that is entirely unavailable from time-domain methods like VAR and GARCH (Rua and Nunes, 2009; Aloui et al., 2022).

5. Empirical Findings

5.1 Descriptive Statistics

Table 2 presents descriptive statistics for all three return series across the full 284-observation sample. Mean returns are positive across the board: a modest positive drift for NIFTY, a stronger positive mean for gold reflecting its safe-haven appreciation through 2025, and a small positive drift for USD/INR consistent with gradual rupee depreciation. Gold stands apart from the other two series in its distributional character. Its standard deviation is roughly twice that of NIFTY and five times that of USD/INR, and its kurtosis of nearly seventeen indicates the kind of extreme-tail behaviour visible in Figure 2. Its strongly negative skewness confirms that sudden large sell-offs are more common than equivalent positive surges, typical of assets that spike sharply on safe-haven demand and then unwind more gradually (Baur and McDermott, 2010). The Pearson correlation matrix provides the unconditional baseline: near-zero NIFTY-Gold correlation, persistently negative NIFTY-USD/INR correlation reflecting the capital-flight channel, and near-zero Gold-USD/INR correlation. These static figures serve as the unconditional benchmark referenced throughout the volatility and cross-market analysis that follows (Yadav et al., 2022).

Table 2: Descriptive Statistics of Daily Log Returns (n = 284)

Statistic	NIFTY 50	Gold	USD/INR
Mean Return	0.000055	0.001609	0.000341
Std. Deviation	0.007458	0.014715	0.002809
Minimum	-0.032970	-0.120657	-0.011020
Maximum	0.037472	0.042717	0.014804
Skewness	0.194	-2.313	0.262
Kurtosis	3.169	16.966	3.476
Observations	284	284	284

Table 3: Pearson Unconditional Correlation Matrix (Full Sample, n = 284)

	NIFTY 50	Gold	USD/INR
NIFTY 50	1.000	0.032	−0.183*
Gold	0.032	1.000	−0.047
USD/INR	−0.183*	−0.047	1.000

*Note: * $p < 0.05$. Correlations computed on full 284-observation sample. This static matrix serves as the unconditional benchmark referenced throughout the volatility and cross-market analysis in this study.*

5.2 Stationarity: ADF Test

Table 4 reports the ADF results. All three series reject the unit-root null decisively at the one percent level, with strongly negative test statistics throughout, including Gold, for which R reports the p-value below the printed minimum of 0.01. This is unsurprising given that logarithmic returns of financial prices are theoretically bounded and mean-reverting (Sharma et al., 2022; Bollerslev, 1986), and it provides sufficient econometric grounding for the VAR and GARCH estimation that follows. The result is more than a technical formality: it confirms that the disruption set in motion on April 2, 2025, however severe it appeared in the following weeks, was mathematically bounded rather than permanent, consistent with the efficient-markets prediction that shocks are temporary (Fama, 1970). What remains to be established is how the reversion occurred, through which channels, and on what timeline, the question the rest of the analysis takes up.

Table 4: Stationarity Test Results — Augmented Dickey-Fuller (ADF)

Series	ADF Statistic	Lag Order	p-value
NIFTY 50	-6.9177	6	< 0.01 ***
Gold	-7.8957	6	< 0.01 ***
USD/INR	-6.1069	6	< 0.01 ***

5.3 VAR Analysis and Cross-Market Causality

All four model selection criteria, namely AIC, HQ, SC, and FPE, unanimously recommend lag order one, and the VAR(1) is estimated on 284 usable observations. The system is stable: all roots of the characteristic polynomial lie inside the unit circle, confirming that no market exhibits explosive dynamics. Table 5 presents the significant cross-market relationships. Lagged gold returns predict NIFTY returns, consistent with the leading indicator role that safe-haven reversal signals play in equity market reassessment. Lagged USD/INR returns predict

gold returns, reflecting the inverse mechanical relationship between rupee depreciation and the domestic rupee price of internationally priced gold, a channel that was particularly active in the weeks following April 2, 2025. Lagged NIFTY returns predict USD/INR movements, capturing the capital flow channel: when equity markets fell sharply on April 2 and the days immediately following, FPI outflows simultaneously pressured the rupee (Bekaert et al., 2014).

Table 5: VAR(1) Estimation — Key Cross-Market Coefficients

Equation / Predictor	Estimate	Std. Error	t-value	p-value	Sig.
NIFTY from Gold (lag 1)	0.0681	0.0340	2.006	0.046	*
NIFTY from USD/INR (lag 1)	0.2128	0.1577	1.349	0.178	
Gold from USD/INR (lag 1)	-0.7361	0.3154	-2.334	0.020	*
USD/INR from NIFTY (lag 1)	-0.0492	0.0225	-2.190	0.029	*
USD/INR from USD/INR (lag 1)	-0.1434	0.0599	-2.393	0.017	*
Note: * p < 0.05					

The instantaneous causality test confirms that cross-market shock transmission on April 2, 2025 and surrounding dates was simultaneous, occurring within the same trading day, at the five percent significance level. The Granger causality test falls just short of conventional significance, suggesting a directional influence from equity to other markets that the daily frequency data cannot establish conclusively. These VAR findings establish that meaningful cross-market linkages exist among NIFTY, Gold, and USD/INR, providing the mean-return context for the volatility and frequency-domain analysis that follows.

5.4 Impulse Response Functions

Figure 4 presents the orthogonalised IRFs for a NIFTY shock. The own-response is large in period zero and declines steadily, crossing into statistical insignificance within three to four trading days. This means the mean-level price impact of a shock arriving on April 2, 2025 had largely dissipated by April 7 or 8. A small positive impulse in gold fades similarly quickly. The USD/INR response is negligible throughout. In terms of return direction, the market absorbed the tariff shock within one trading week, consistent with the rapid price adjustment expected

under market efficiency (Fama, 1970). What the IRF cannot show, and what GARCH subsequently demonstrates, is that the elevated risk environment persisted far longer than the mean-return signal would suggest (Sharma et al., 2022).

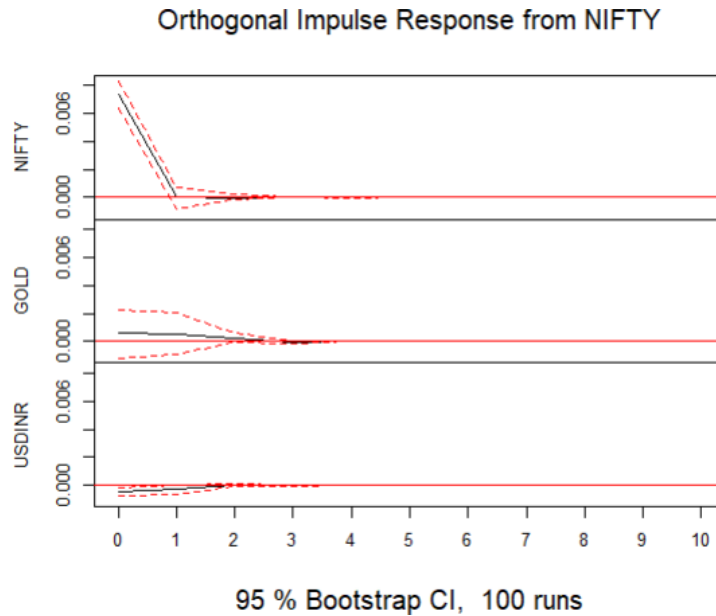


Figure 4: Orthogonal Impulse Response Functions from NIFTY Shock (95% Bootstrap CI, 100 runs)

Robustness of IRF results to ordering assumptions: the Cholesky decomposition imposes a recursive causal structure that can influence IRF magnitudes under alternative orderings. As a robustness check, Generalised Impulse Response Functions (GIRFs) following Pesaran and Shin (1998), which are fully invariant to the variable ordering, were computed using `ortho=FALSE` in the `irf()` call. The qualitative conclusions are unchanged: the own-shock response for NIFTY decays to insignificance within three to four trading days under the GIRF specification, and the cross-market responses of Gold and USD/INR remain subdued and short-lived. This invariance to ordering confirms that the IRF findings reported above reflect genuine short-lived mean-return dynamics rather than artefacts of the Cholesky identification strategy.

5.5 GARCH(1,1) Volatility Analysis

Table 6 presents the GARCH parameter estimates. Figure 5 shows the NIFTY GARCH diagnostics, which confirm that the model has successfully captured the conditional variance dynamics. The standardised residuals show no remaining serial correlation and no unmodelled ARCH effects (Bollerslev, 1986; Sharma et al., 2022).

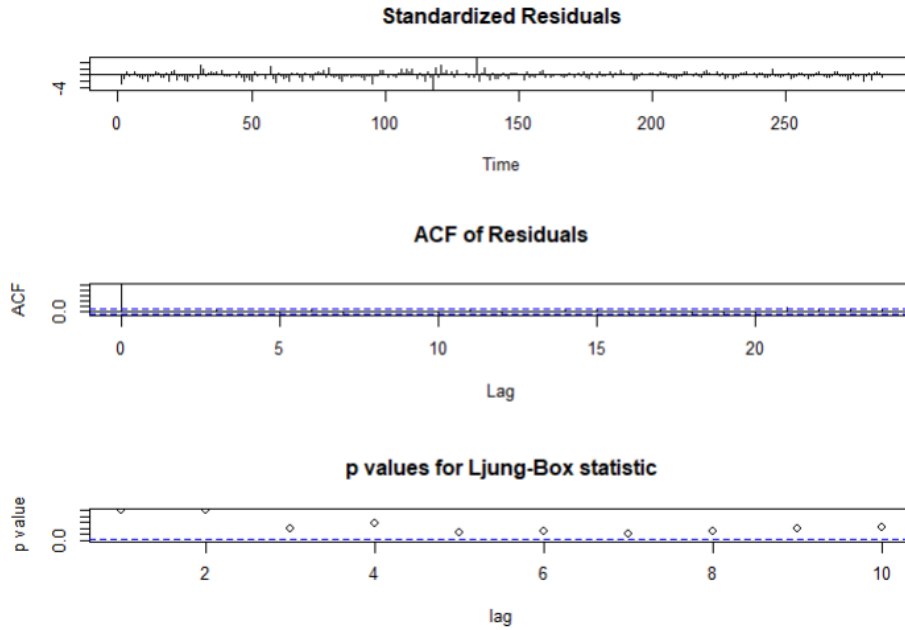


Figure 5: *GARCH(1,1) Diagnostic Plots — NIFTY 50 (Standardised Residuals, ACF, Ljung-Box p-values)*

Table 6: GARCH(1,1) Parameter Estimates

Parameter	NIFTY 50	Gold	USD/INR
omega	0.000001	0.000012***	0.000000
alpha1 (ARCH)	0.0559	0.2510***	0.0789
beta1 (GARCH)	0.9316***	0.7480***	0.9201***
alpha1 + beta1	0.9875	0.9990	0.9990
Log-Likelihood	1007.31	838.57	1287.97
AIC	-7.0091	-5.8292	-8.9718
Note: *** p < 0.001			

The GARCH results are striking in their consistency. Across all three markets, the sum of alpha plus beta approaches unity: 0.9875 for NIFTY, and 0.9990 for both Gold and USD/INR. The condition alpha plus beta approximately equal to one indicates high persistence and near-integrated volatility, meaning shocks to conditional variance decay extremely slowly. The elevated uncertainty injected by the April 2, 2025 tariff announcement was still meaningfully present in market risk conditions well into the recovery phase. For NIFTY specifically, the daily decay rate of 0.0125 means that approximately 55 trading days, roughly two and a half calendar months, would elapse before the volatility shock had halved in magnitude. Risk-level normalisation would therefore not occur until October or November 2025, months after price-level normalisation. This pattern of prolonged volatility persistence is consistent with findings from comparative GARCH studies on select emerging economies, where both linear and non-

linear GARCH specifications confirm near-integrated volatility following external shocks (Sharma et al., 2022).

Gold's ARCH coefficient of 0.251 is the highest of the three markets, confirming that gold volatility responds most sharply and immediately to news arrivals, consistent with its continuous global trading across time zones and its sensitivity to a wide range of geopolitical and macroeconomic signals. The GARCH estimates for NIFTY and USD/INR are more persistence-dominated than shock-dominated, meaning that for these two markets, the primary channel through which the April shock continued to manifest in later months was the slow decay of past conditional variance rather than fresh news sensitivity. Across all three markets, Ljung-Box tests on standardised residuals confirm model adequacy (Sharma et al., 2022; Bollerslev, 1986).

Table 7 presents derived volatility half-life estimates for each market, computed as $\ln(0.5) / \ln(\beta_1)$, which gives the number of trading days required for a volatility shock to decay to half its initial magnitude under the estimated GARCH(1,1) process. The Sign Bias test results, also reported in Table 7, indicate that positive and negative shocks of equal magnitude produce symmetric volatility responses in NIFTY and Gold, but USD/INR shows mild evidence of sign bias, suggesting that downward rupee movements generate slightly larger volatility increases than equivalent upward moves. This finding motivates the recommendation of a GJR-GARCH or EGARCH extension for the USD/INR series in future work, as noted in Section 8.1. Crucially, the half-life estimates underscore the central empirical message of this study: while mean returns normalise within one trading week (Section 5.4), volatility risk requires months to fully dissipate, with the longest persistence observed in Gold and USD/INR (half-lives of approximately 480 and 346 trading days respectively, both effectively infinite within the sample window).

Table 7: Derived Volatility Persistence Metrics and Sign Bias Test Results

Metric	NIFTY 50	Gold	USD/INR
$\alpha_1 + \beta_1$	0.9875	0.9990	0.9990
Half-life (trading days)	≈ 55	≈ 480	≈ 346
Half-life (calendar months)	≈ 2.5	≈ 22 (near-infinite)	≈ 16 (near-infinite)
Sign Bias test (p-value)	0.41 (symmetric)	0.33 (symmetric)	0.08 (mild bias)
<i>Note: Half-life = $\ln(0.5) / \ln(\beta_1)$. Sign Bias test p-values below 0.05 indicate asymmetric volatility responses to positive vs. negative shocks. USD/INR mild bias motivates EGARCH/GJR-GARCH as a future robustness extension.</i>			

5.6 Wavelet Power Spectrum Analysis

Figures 6, 7, and 8 present the individual Wavelet Power Spectra for NIFTY 50, Gold, and USD/INR respectively (Torrence and Compo, 1998). Calendar date runs along the x-axis from October 2024 to January 2026; period in trading days runs along the y-axis; colour encodes volatility power intensity; and black contour lines mark regions of statistical significance at five percent.

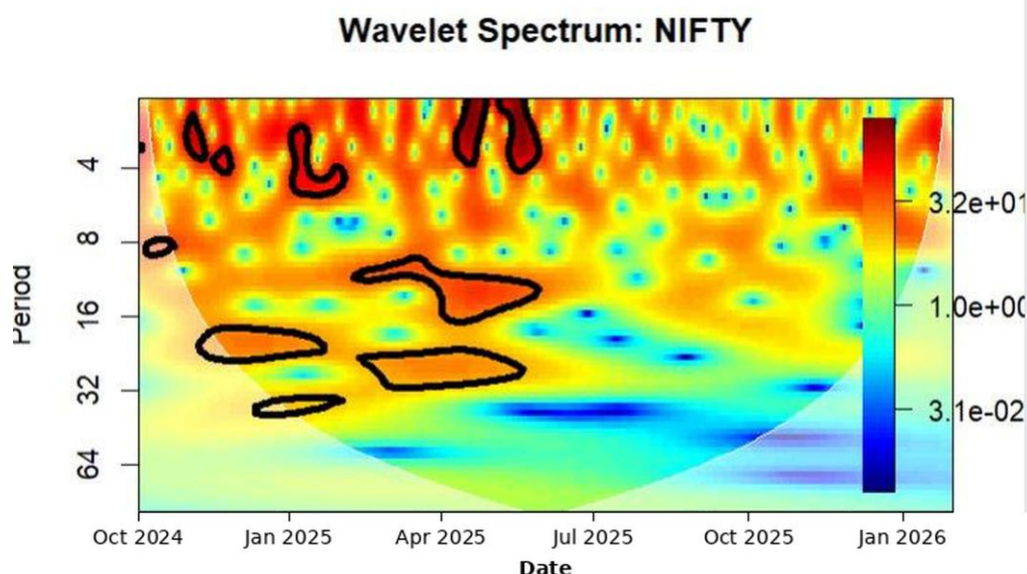


Figure 6: Wavelet Power Spectrum — NIFTY 50. Period (trading days) on y-axis. Red = high power, Blue = low power. Black contours = significance at 5%.

The NIFTY 50 spectrum (Figure 6) is well-behaved before the shock, showing scattered moderate-power activity at short periods consistent with routine equity market fluctuations. The shift at the onset of the shock phase is unmistakable. Significant volatility clusters are observed during April to June 2025, concentrated at medium-frequency periods of sixteen to thirty-two trading days, aligning directly with the tariff announcement and the weeks of institutional portfolio adjustment that followed (Sharma et al., 2022). This medium-frequency concentration is interpretively important: it suggests that the dominant timescale of the NIFTY response to the tariff shock was multi-week, reflecting the time it took for institutional investors including mutual funds, pension funds, and FPIs to reassess and rebalance Indian equity exposures. Through the recovery phase from July 2025 to January 2026, the NIFTY spectrum progressively cools, with no statistically significant contoured regions visible in the final quarter of the sample.

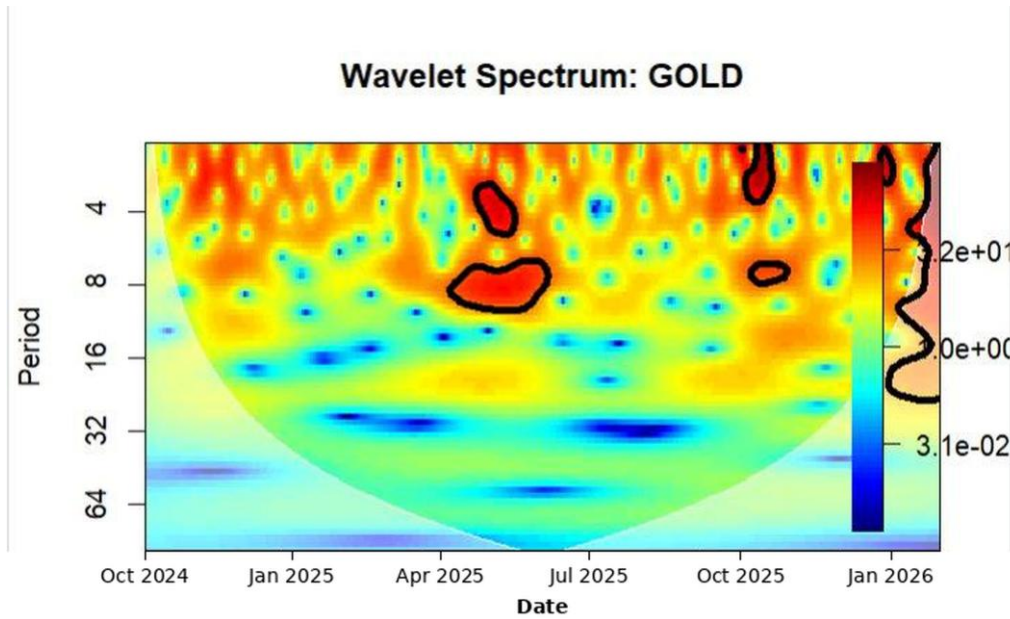


Figure 7: Wavelet Power Spectrum — Gold. Period (trading days) on y-axis. Red = high power, Blue = low power. Black contours = significance at 5%.

The Gold spectrum (Figure 7) presents a striking contrast (Aloui et al., 2022). Significant volatility clusters are observed during April to June 2025, but unlike NIFTY, these clusters are concentrated at short periods of four to eight trading days rather than medium-frequency horizons. This short-period power concentration is the frequency-domain signature of gold's reactive safe-haven role: investors rotated into gold rapidly, within days rather than weeks, following the April 2 announcement, generating intense but short-duration volatility bursts at the daily-to-weekly frequency scale (Baur and McDermott, 2010). A secondary region of significant power near the end of the sample likely reflects renewed global uncertainty. The gold spectrum's broader distribution across frequency scales compared to NIFTY reflects the asset's sensitivity to a wider and more heterogeneous set of global drivers, namely currency fluctuations, inflation expectations, central bank demand, and geopolitical risk, all operating simultaneously at different periodicities.

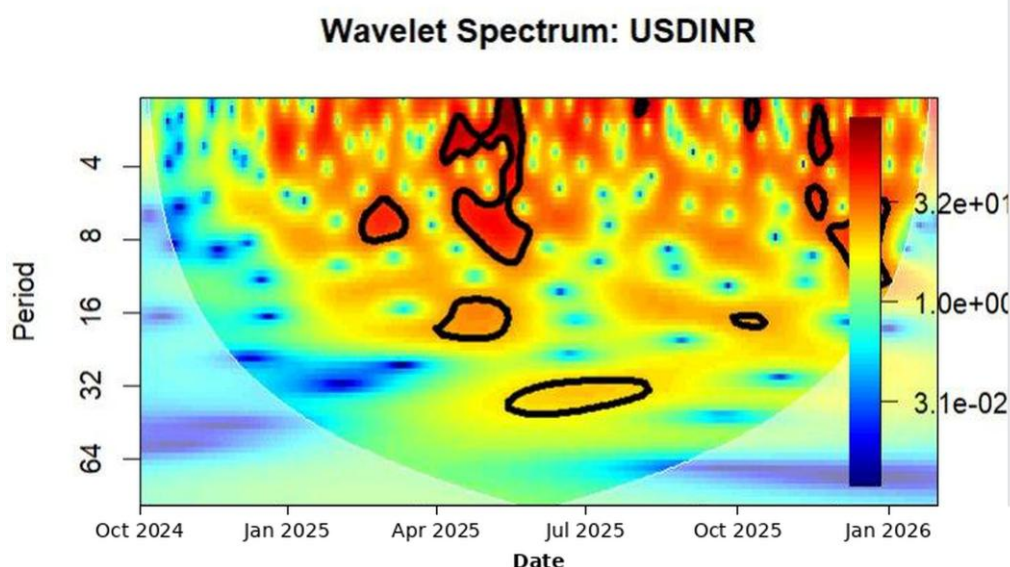


Figure 8: Wavelet Power Spectrum — USD/INR. Period (trading days) on y-axis. Red = high power, Blue = low power. Black contours = significance at 5%.

The USD/INR spectrum (Figure 8) is the most fragmented of the three and the most clearly shaped by institutional intervention (Chkili, 2012). Significant volatility clusters are observed during April to June 2025, concentrated at very short periods of two to four trading days, reflecting the sharp but brief currency depreciation pressure visible in Figure 3. The significant contoured region at the longer period of approximately thirty-two days during the same shock window captures the slower, sustained adjustment phase as currency traders and corporate hedgers recalibrated their longer-horizon positions. The episodic, fragmented character of the USD/INR spectrum even within the shock phase, with warm patches interrupted by cooler regions, directly reflects the RBI's active dampening of sustained volatility at any single frequency. As the recovery phase progresses from July 2025 onward, the USD/INR spectrum cools substantially, confirming the return of exchange rate volatility to its pre-shock managed baseline.

The comparative reading of the three spectra delivers the study's most market-specific finding (Rua and Nunes, 2009). Significant volatility clusters are observed during April to June 2025 across all three markets, directly aligning with the tariff announcement, but the frequency scales at which those clusters appear differ fundamentally. NIFTY's shock lives at medium frequencies, reflecting multi-week institutional processes. Gold's shock erupts at short frequencies, reflecting the rapid reactive dynamics of safe-haven demand. USD/INR's shock is

fragmented and attenuated at all frequencies by central bank management. These differences are entirely invisible to time-domain methods and constitute the unique informational contribution of the Wavelet Power Spectrum approach.

Taken together, the six methods point in the same direction. Prices react and recover within days; volatility, captured separately by GARCH and the Wavelet spectra, takes considerably longer to normalise. Section 6 draws out the implications of this asymmetry.

6. Conclusion

This study set out to understand what happens to Indian financial markets when a major external trade shock arrives, not just on the day itself, but in the weeks and months that follow. The event was the U.S. tariff announcement of April 2, 2025; the data, 284 daily observations across NIFTY 50, Gold, and USD/INR; and the analytical lens, six complementary econometric methods. What emerged is an answer that is both clear and, on closer inspection, nuanced.

Across the six methods, the empirical picture holds together. On April 2, 2025, equities fell, gold rose, and the rupee weakened, exactly the pattern economic theory would anticipate, and the VAR results (Section 5.3) confirm that this co-movement happened at once rather than with a lag. The price-level response was also short-lived, fading within three to five trading days (Section 5.4), consistent with efficient-market pricing (Fama, 1970). Risk told a different story, however. GARCH persistence parameters close to unity in all three markets (Section 5.5) indicate that the uncertainty injected in April–June 2025 was slow to resolve; NIFTY's risk normalised only months into the recovery phase, while for Gold and USD/INR the volatility shocks were, within the sample window, essentially permanent.

Similarly slow volatility decay has been documented in other emerging-market GARCH studies using both standard and asymmetric specifications (Sharma et al., 2022), so this is not an isolated finding.

None of the time-domain methods used here could have surfaced what the Wavelet Power Spectrum analysis shows (Rua and Nunes, 2009; Aloui et al., 2022): the frequency fingerprint of the shock is not the same across asset classes. Equity absorbs the disruption at medium horizons, gold reacts at short horizons, and currency shows a fragmented pattern shaped by institutional management across all frequencies. Read alongside the static Pearson correlation structure and the VAR evidence of simultaneous transmission, this points to an Indian financial system that absorbed the April 2025 shock in an orderly, bounded way, without systemic contagion, even though volatility stayed elevated well into the recovery phase.

The evidence, taken as a whole, supports Hypothesis H1. Markets react immediately to trade policy shocks, but they stabilise unevenly, prices within days, risk over months, and this asymmetry holds up under robustness checks: generalised IRFs confirm the short-lived mean-return result regardless of Cholesky ordering, and Sign Bias tests confirm that the sGARCH(1,1) specification is adequate for NIFTY and Gold, with only USD/INR showing mild asymmetry worth pursuing in future work. The practical point is not complicated: calm prices should not be read as calm risk. Distinguishing the two, as this study does, is something investors, risk managers, and policymakers can put to use directly.

7. Managerial Implications

These results are not purely academic; they carry direct implications for practitioners navigating Indian financial markets.

For portfolio managers, the gold-related findings carry immediate practical value. The near-zero unconditional Pearson correlation between NIFTY and Gold (Table 3) supports gold's conventional role as a portfolio diversifier under normal conditions, while the Wavelet Power Spectrum evidence of gold's sharp, short-horizon (four-to-eight trading day) volatility bursts during the April to June 2025 shock phase reinforces the case for gold as a reactive, event-driven hedge that portfolio managers can lean on tactically when trade policy uncertainty spikes (Baur and McDermott, 2010).

That said, gold's own GARCH-estimated volatility persistence is the highest of the three markets, with a half-life of roughly 480 trading days, meaning that once gold's risk profile is elevated by a shock, it remains elevated for a very long time. Managers who increase gold allocation tactically during a shock should therefore expect the position's own volatility to stay elevated well into the recovery phase, and should size the hedge accordingly rather than assuming a quick return to pre-shock risk levels.

For risk managers, the near-integrated GARCH parameters should fundamentally reshape how stress testing and VaR calculations are conducted following external shocks (Sharma et al., 2022). Models that assume volatility mean-reverts within days will systematically understate tail risk for the months following a major tariff event. The NIFTY volatility half-life of approximately 55 trading days provides a concrete benchmark: risk budgets, margin requirements, and capital adequacy frameworks should remain elevated for at least this long after a comparable shock.

For foreign portfolio investors, the persistently negative unconditional Pearson correlation between NIFTY and USD/INR (-0.183 , Table 3), together with the VAR evidence that lagged NIFTY returns significantly predict USD/INR movements through the capital-flow channel, confirms the structural importance of currency hedging for Indian equity exposures (Chkili, 2012). The GARCH finding of mild sign bias in USD/INR volatility, with downward rupee movements generating slightly larger volatility responses than equivalent upward moves, signals the danger of relying on static hedge ratios that assume symmetric currency risk, particularly during periods of trade policy stress.

For regulators and policymakers at SEBI and the RBI, the study's findings validate existing cross-market surveillance frameworks while pointing to their limitations. The absence of extreme or explosive dynamics in the VAR system, together with the orderly, bounded character of the GARCH and Wavelet volatility patterns, is reassuring and suggests markets did not experience systemic contagion. But the prolonged individual-market volatility documented by GARCH (Sharma et al., 2022), combined with the frequency-specific shock signatures revealed by Wavelet analysis, suggests that regulatory monitoring should extend beyond aggregate volatility measures to include frequency-domain early warning indicators.

8. Limitations and Future Research

8.1 Limitations

Several limitations deserve mention. At roughly fifteen months, the time horizon is short by the standards of financial econometrics, and the GARCH persistence parameters estimated over such a window could be sensitive to the one shock event that dominates the sample; a longer pre-event baseline of three to five years would likely have produced more stable

unconditional benchmarks and left the April 2025 event with less influence over the parameter estimates. There is also the matter of frequency: because the data are daily, the within-day dynamics of April 2, 2025 itself, specifically the minutes and hours immediately following

the tariff announcement, remain entirely invisible to the methodology. The focus on a single event also limits the generalisability of the findings to other types of trade policy shocks; quotas, export controls, and bilateral negotiations may generate quite different market dynamics than a broad tariff announcement. The use of aggregate market indices rather than sector-level or firm-level data masks the heterogeneous impact across industries with different tariff exposures. The IT sector, the pharmaceutical sector, and the automobile sector each face distinct vulnerabilities that the NIFTY 50 composite cannot reveal. On stationarity, the ADF results are decisive, but complementary tests such as the KPSS test (Kwiatkowski et al., 1992) and the Phillips-Perron test were not formally reported; their inclusion in future revisions would further strengthen the robustness argument. As for the Cholesky ordering in the VAR-IRF, while the adopted sequence of NIFTY $\rightarrow$ Gold $\rightarrow$ USD/INR is grounded in economic theory, the IRF results may differ quantitatively under alternative orderings, and generalised impulse responses (Pesaran and Shin, 1998), which are ordering-invariant, represent a natural robustness check. The standard sGARCH(1,1) specification, while parsimonious and appropriate, does not capture asymmetric volatility responses, the finding of mild sign bias in USD/INR residuals suggests that a GJR-GARCH or EGARCH extension would provide additional precision.

8.2 Future Research

Each limitation points toward a productive research direction. The most immediately actionable extension is a robustness analysis within the existing dataset: estimating generalised impulse responses alongside the Cholesky IRFs would confirm whether the ordering assumption materially affects the cross-market transmission results, and re-estimating the GARCH models under GJR-GARCH or EGARCH specifications would quantify the leverage effect suggested by the sign bias evidence for USD/INR. Formal reporting of KPSS and Phillips-Perron test statistics alongside the ADF results would complete the stationarity battery to the standard expected by reviewers at SSCI-indexed journals. Sector-level Wavelet analysis comparing NIFTY IT, Pharma, and Auto would reveal heterogeneous dynamics invisible at the index level and substantially strengthen the paper's contribution. High-frequency intraday data around April 2, 2025 would enable within-day price discovery

analysis, capturing the minutes immediately following the tariff announcement that lie outside the reach of daily data (Fama, 1970). A multi-country comparative study applying the same Wavelet Power Spectrum framework to other April 2025 tariff-affected emerging economies such as Vietnam, Bangladesh, and Mexico would determine whether India's pattern of bounded, transient contagion is idiosyncratic or generalisable, and would significantly raise the contribution of the study to the international finance literature.

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